

Use-Case Flow Specification

Use Case: Customize Monthly Box **ID:** UC-002 **Primary Actor:** Subscriber

Secondary Actor(s)	System (Recommendation Engine, Fulfillment Service)
Related Requirements	FR-001, FR-002, FR-005
Description	Allows a subscriber to swap, add, or remove items in the upcoming monthly box based on personal preferences, provided the customization window is still open (≥ 48 hours before billing renewal).

Preconditions

- Subscriber is registered and has an active (non-cancelled) subscription.
- Subscriber is authenticated and logged into the portal.
- Next billing cycle is scheduled, and current date/time is at least 48 hours before the monthly billing renewal date.
- A default box has been generated for the upcoming cycle based on the subscriber's preference tags.

Postconditions

- Monthly box contents are updated to reflect the new selections.
- Fulfillment manifest is refreshed with the updated item list.
- Confirmation notification (email + in-app) is sent summarizing the changes.
- Inventory for swapped-out items is restored and swapped-in items are reserved.

Main Success Scenario

Step	Actor	Action
1	Subscriber	Navigates to the "My Upcoming Box" page from the dashboard.
2	System	«include» Validate Customization Window: Checks current time is ≥ 48 hours before billing renewal. Validation passes.
3	System	«include» Apply Preference-Based Recommendations: Loads current box contents alongside personalized recommendations based on preference tags.
4	System	Displays current box items, alternative products with preference-match scores, and customization deadline.
5	Subscriber	Reviews the current box and selects one or more items to swap, choosing replacements from recommendations or the full catalog.
6	System	Validates each replacement is in stock and compatible with the subscriber's subscription tier.
7	System	Updates the box preview in real time with new items, adjusted summary, and any price difference.
8	Subscriber	Reviews the updated preview and confirms changes by clicking "Save My Box."
9	System	Persists updated selections; releases holds on removed items and reserves inventory for newly selected items.
10	System	Refreshes the fulfillment manifest for the current billing cycle.
11	System	Sends confirmation notification (email + in-app) with finalized box summary and expected delivery date.
12	System	Displays: "Your box has been updated successfully!"

Alternate Flow: Customization Window Closed

Step	Actor	Action
2a	System	At Step 2, determines current time is less than 48 hours before billing renewal; customization window has closed.
2b	System	Displays: "The customization window for this cycle has closed. Your current box will be shipped as-is. You can still customize your next month's box after the current cycle renews."
2c	System	Shows current (locked) box contents in read-only mode and the date when customization for the next cycle will open.
2d	Subscriber	Acknowledges the message; may navigate to subscription settings to skip the current cycle if the skip window is still open, or simply exit.